

26. (1 point)

An insurance company with a substantial book of long-tailed business is using a plan loss ratio model to determine premium growth targets. The plan loss ratio is also used in the reserve review process as the expected loss ratio.

a. (0.5 point)

Identify two potential negative consequences of an optimistic plan loss ratio to the company's financial results.

b. (0.5 point)

Explain why it is difficult to separate operational risk from underwriting risk when explaining the impact of an optimistic plan loss ratio on the company's financial results in retrospect.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 26	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: C7: Describe operational risk and demonstrate possible mitigation and quantification methodology.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answers, any two non-overlapping ones of which would earn full credit</i> • Optimistic planned loss ratio use as ELR can lead to reserve deficiencies. • Optimistic planned loss ratio use as ELR can lead to inadequate reserve. • Optimistic planned loss ratio use as ELR can lead to reserve conflagration. • Optimistic planned loss ratio use as ELR can lead to premium growth in a line that is not as profitable as previously thought. • Optimistic planned loss ratio use as ELR can lead to underpricing business. • Optimistic planned loss ratio use as ELR can lead to rating downgrade. • Optimistic planned loss ratio use as ELR can lead to policyholder exodus. • Optimistic planned loss ratio use as ELR can lead to claim paying difficulty. • Optimistic planned loss ratio use as ELR can lead to insolvency issue. • Optimistic planned loss ratio use as ELR can lead to suboptimal investment strategy. • Optimistic planned loss ratio use as ELR can lead to investors' dissatisfaction.	
Part b: 0.5 point	
<i>Sample Answer 1</i> Operational risk could have been the cause of the optimistic loss ratio – an inadequate review process, perhaps, or a deficient unpaid loss estimation algorithm, or management pressured the actuaries to select more optimistically than they would have otherwise. The model's inability to accurately forecast the loss ratio would be a manifestation of underwriting risk. It may be difficult to objectively determine the level of accuracy in a model's ability to forecast a loss ratio, making it difficult to distinguish underwriting risk from operational risk. <i>Sample Answer 2</i> It is hard to tell if the forecasting model could not predict the loss ratios or was not used appropriately. If forecasting model couldn't predict accurately, it is underwriting risk (if other companies are facing the same problem). But if the model the model was not used appropriately, it is operational risk. <i>Sample Answer 3</i> Underwriting risk incorporates the random volatility inherent in insurance losses, operational risk incorporates the inadequate or failed internal processes and people. You could argue that the LR deterioration is due to underwriting risk that could not have been modeled or alternatively that the models were not appropriately used resulting in operational risk.	
EXAMINER'S REPORT	
Part a	
A wide range of alternative answers could earn full credit. Candidates were expected to identify two distinct potential issues.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Most candidates did well and received full credit on part a. The common errors made by candidates including repetitive answers (e.g., "Under-reserving in most recent AY" and "Under-reserving for all prior AYs").

Part b

Candidates were expected to explain the overlap between operational and underwriting risk when discussing how a loss ratio selection has played out.

Few candidates earned full credit. Common errors included:

- Not explaining underwriting or operational risk
- Not labeling which risk is illustrated by the example candidates provided
- No statement explaining why it is difficult to separate the two risks
- Simply reiterating that it is difficult to separate the two risks